

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461091.6292 +/- 0.0081 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.02
Transit depth (Rp/Rs)^2	1.08 +/- 0.42 [%]
Orbital Inclination (inc)	89.9 +/- 1.54
Airmass coefficient 1 (a1)	0.8793 +/- 0.005
Airmass coefficient 2 (a2)	0.09 +/- 0.19
Scatter in the residuals of the lightcurve fit is	0.57 %
Best Comparison Star	#3 - [221, 227]
Optimal Aperture	3.18
Optimal Annulus	12.68
Transit Duration (day)	0.0929 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.104 ± 0.02 vs 0.11 (deviation 0.3σ)
Duration fit vs published	0.0929 d vs 0.1208 d (deviation 3.29σ)
Mid-time O–C	-15.3 min
Depth SNR	5.2
Residual scatter	0.57 %

Light curve

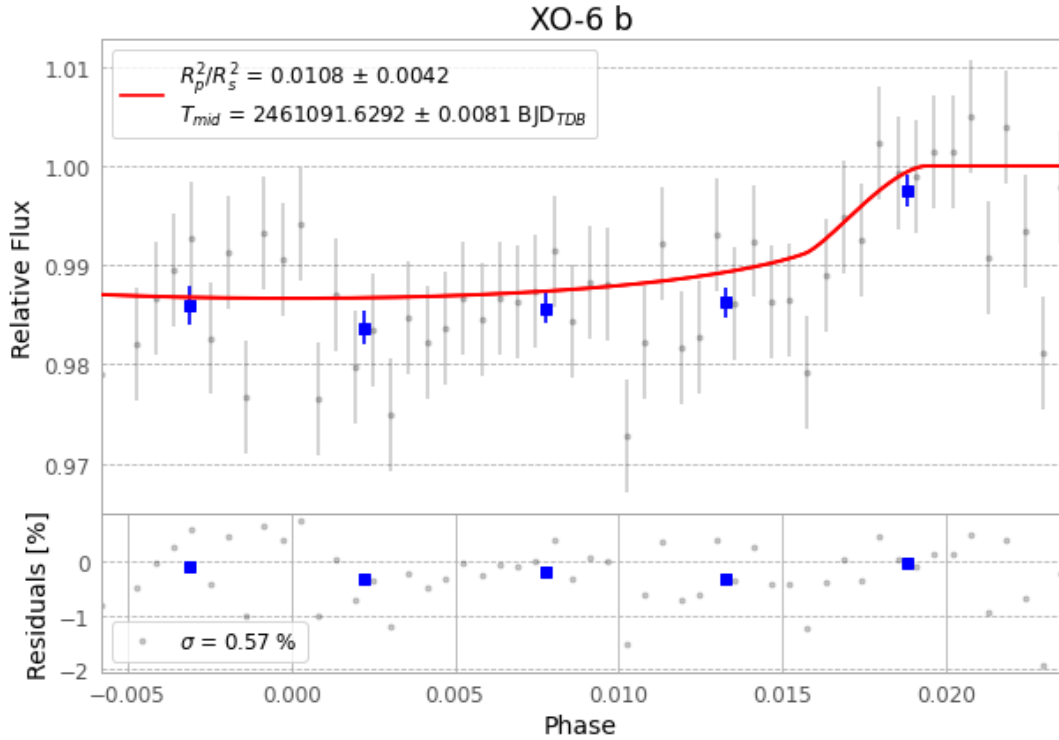


Figure 1 — FinalLightCurve_XO-6 b_2026-02-19.png (SHA-256 dd3f27c9f2961a5d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 83], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 dc38b705aab846a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

53 FITS frames (35 MB), XO-6260220023014.FITS → XO-6260220050915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-260220.log (a30f11b8881b...), FinalLightCurve_XO-6 b_2026-02-19.png (dd3f27c9f296...), FinalLightCurve_XO-6 b_2026-02-19.csv (b8acdeb462d1...)

Compute resources

Wall time 924.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:17Z.